Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/19/2006
PI Name:	Thompson Jr, James N. Ph.D.		
Project Title:	Genetic and Developmental Stability in Response to Long-Term Exposure of Drosophila melanogaster to a Space Station Environment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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Zip Code:	73019-3032	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Space Biology ILSRA
Start Date:	08/01/2002	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	Currently in Phase B for feasibility and strain characterization studies. NOTE: NCE to 9/30/2007, per A Chu 7/2006.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Woodruff, Ronny (Bowling Green State University)		
Grant/Contract No.:	NCC2-1355		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Using a simple invertebrate model, the fruit fly Drosophila melanogaster, we will study the effect of the space environment on: (1) successful reproduction over a period of seven generations on the International Space Station (ISS) in microgravity and 1 g control conditions; (2) genetic stability as measured by mutation accumulation during this multi-generation period; (3) the effect of microgravity, radiation, and other elements of the space environment on predicted reductions in developmental homeostasis (regulatory buffering) in response to environmental and/or mutational stresses; and (4) protein, DNA mutation, or RNA profiles from which stress mechanisms can be analyzed.. Mutation accumulation will be measured in genetic loci that cause an altered lifespan, that reduce viability (i.e., reduce relative fitness), that contribute to increased heritable variation in neurosensory organ number (principally sternopleural bristle number), and that yield DNA microsatellite and sequence variation. Changes in expression of genetic repair and other stress-related genes by proteomic profiles, RNA microarray, or comparable techniques is also of interest. The primary morphological characters assayed for developmental stability will be changes in the pattern of wing support vein patterns and changes in the number and distribution of surface sensory organs (e.g., bristles and the campaniform sensillae located on the third longitudinal [L3] vein). A deviation from symmetry (fluctuating asymmetry, FA) measures the quality of developmental regulation in traits that are normally symmetrical. Together, these measures of genetic and developmental stability will extend our understanding of the rate and types of genetic changes that can occur during long-term exposure to a space environment and the effects of this novel stress on developmental homeostasis. This helps answer key questions about the ability of biological systems common to all organisms to adapt to novel environmental conditions of a space environment. The scientific goals for the third year (Phase C) focus on the characterization of breeding programs and prospective strains of Drosophila melanogaster for planned implementation of a multi-generation mutation accumulation experiment on the International Space Station. As the hardware is being developed and tested for this ISS experiment, the goals are to: (1) continue to characterize selected strains for mutation sensitivity, sequence DNA at targeted sites, carry out microsatellite allele profiles, and measure baseline fluctuating asymmetry (FA) of morphological markers; (2) initiate RNA stability tests and evaluate alternative methods for maintaining stable DNA, RNA and protein samples under conditions available on the ISS; and (3) further characterize the Binscy Assay for mutation accumulation over multiple generations as a basis for later evaluation of biocompatibility tests in the ISS habitats. Exposure to the novel combination of physical stresses in a space environment can have both adverse and potentially beneficial effects on genetic stability and stress response pathways. Experiments with characterized genotypes of the model organism Drosophila melanogaster will provide important insights into conserved biological responses to novel environmental stress.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of a long-term exposure to the radiation and gravitational condiditions of a space environment will extend our knowledge of the sensitivity and responsiveness of genetic and developmental systems to novel stresses. This information will help assess measures required for successful colonization and utilization of space.
Task Progress:	This year's focus was on testing biological parameters of the proposed breeding programs in newly designed chambers and characterizing strains for a multiple generation mutation accumulation experiment on the International Space Station. During this period, it became clear that resources no longer could support the initial experimental plans. This led us to review and modify the experimental requirements for our experiment with Drosophila melanogaster on the International Space Station As a result, a number of options have been developed. Since a mutation accumulation assay requires multiple generations of exposure to the space environment conditions, such as microgravity, the highest priority is given to a 90-day experiment, although a 60-day or even 30-day experiment would yield valuable information. Experiment strategy alternatives include: (1) eliminate the need for extra Drosophila Chambers (DCs) by redesigning the experiment into multiples of a 15-day life cycle and fixing the number of cycles rather than allowing it to vary; (2) reduce culture temperature of final cycle to increase scheduling flexibility; (3) reduce astronaut time required for sampling; (4) reduce hazard associated with fixatives by evaluating alternative storage procedures that have recently become available; and (5) possibly automating generation transfers to save astronaut time requirements. This plan was formally submitted in April 2005. Regrettably, in mid-December 2005 we were informed that Drosophila experiments aboard the ISS were being cancelled, with a project ending date of 30 September 2006. The final months are being used to complete the characterization of the strains, complete testing a model for DNA microsatellite mutation rate estimates in vibration and hypergravity, and continue and analyze experiments currently underway to study the effects of short duration vibration, hypergravity, and possibly radiation exposure in stimulating a protective repair and stress response. Recent results from these latter experiments are summarized here. Fluctuating asymmetry (FA) and developmental stability under hypergravity, vibration, nutritional, and radiation stress: Nutritional stress during critical stages of development forces Drosophila into an earlier developmental trajectory, leading to significantly smaller adults. The results from other studies suggest that this smaller morph may have a higher mutation rate. A higher mutation rate and/or reduced uniformity (homeostasis) of development might facilitate the expression of novel characters and, thus, increased the ability to adapt to new conditions. We are testing the hypothesis that small, nutritionally stressed, Drosophila from this alternative developmental trajectory are less developmentally stable as displayed in wing fluctuating asymmetry. If this is indeed a last-ditch-effort survival strategy, then it is reasonable to hypothesize that the energy requirements of maintaining phenotypic integrity are not quite as important as when development is proceeding under normal (unstressed) conditions. Alternatively, stresses could lead to enhanced resource allocations to developmental quality (a form of "hormesis"). We assayed fluctuating asymmetry under normal and stressed conditions: wild type (Canton-S), the methuselah mutant, and an hsp90 deficient strain, and we gathered initial data on mutation rate in stressed and unstressed adults collected from nature. In contrast to some findings by other researchers, we have not yet detected a difference in mutation rate in comparing large and small flies collected in nature. Our assumption is that fly size is a measure of environmental stress during their developmental period on a natural food source. Two clusters of 2 mutations each were found among the progeny of the Big flies and one cluster of 3 mutations was found from the Little flies (Fisher $P = 0.24$ if clusters are counted as single mutation events). To assay nutritional stress-induced effects on developmental coordination, flies were raised from food bottles containing depleted nutrient levels. Those from the second bi-weekly sample (S2) are significantly smaller than those raised on fresher food (S1) (e.g., Canton-S, wing length on average 15% smaller in S2 than in S1; $P < 0.001$). FA

	is consistently higher (that is, developmental coordination of symmetrical traits is poorer) in stressed S2 Canton-S flies, compared to unstressed S1 flies. This can be seen in the trend for larger values of FA for many of our 12 standard wing shape parameters. Relationships of this type are important for correct interpretation of nutritional versus other stress effects, like hypergravity, vibration, or radiation in a space environment. In contrast, the effects of nutritional stress are not seen in the more stress-resistant methuselah genotype, although this is a distinctly different kind of stress assay than done previously by others. In contrast to our prediction, however, the hsp-deficient strain did not show higher FA levels in small S2 flies. We had expected that the hsp-deficient genotype would be more sensitive to this nutritional stress and that resource allocation for developmental symmetry would be reduced. These results may, however, be consistent with reports that trait symmetry is independent of Hsp90 buffering. In an overall comparison among genotypes, there are significantly higher levels of fluctuating asymmetry in Hsp90-deficient and methuselah genotypes than in the Canton-S wild type (ANOVA, $F_{2,35} = 13.7$, $P < 0.001$). We had predicted that methuselah would have increased symmetry (reduced FA), and these genotypes are being investigated further in an expanded analysis and under physical stress, including hypergravity. The levels of fluctuating asymmetry increased significantly for many wing measures of offspring from females exposed to 5?g hypergravity or modeled shuttle mid-deck vibration. Interestingly, in seven of the eight significant instances (88%) of a significant 5?g times vibration interaction, the level of FA was reduced in the double treatment. Hypergravity was given approximately one hour before the vibration exposure in these experiments. This suggests that exposure to one stress may stimulate a protective stress response that causes developmental stability to be improved when the same individuals are faced with a second stress. FA and mutation accumulation: The Binscy Assay in a radiation environment allows accumulation of mutations on the X chromosome. FA in Binscy males was compared to that of control siblings at generations 0, 4, 8, 12, and 16. Mutation reduces developmental stability; representative examples are shown here. The effect is most clearly seen in wing measures that are most homeostatically expressed (L 1-3, slope = 0.16 in Binscy and 0.06 in C(1;Ys); and L 2-5, slopes = 0.22 vs 0.07). At G16, mutation accumulation Binscy vs C(1;Ys): wing length, $F = 2.71$; wing width, $F = 2.78$ (both $P < 0.001$). Accumulated mutations do not have a detectable effect on wing measures that are more labile (e.g., Length 1-9). Molecular analyses: Molecular analyses and proteomic profiles of vibrational stress effects and stress pathway genes are in progress and preliminary data analyses have not yet been completed. They will be reported in detail in the final report due later this year.

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